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HelloWorld.cpp

Stack Operations (PUSH, POP, PEEK)

AI CPP RUN

```c
#include <stdio.h>
#define MAX 5

int stack[MAX], top = -1;

void push(int value) {
    if (top == MAX - 1)
        printf("Stack Overflow\n");
    else {
        stack[++top] = value;
        printf("Inserted %d\n", value);
    }
}

void pop() {
    if (top == -1)
        printf("Stack Underflow\n");
    else
        printf("Deleted %d\n", stack[top--]);
}

void peek() {
    if (top == -1)
        printf("Stack is Empty\n");
    else
        printf("Top element is %d\n", stack[top]);
}

void display() {
    for (int i = top; i >= 0; i--)
```

STDIN

Input for the program (Optional)

Output: 3 ms | 4.4 MB

```
Inserted 10
Inserted 20
Inserted 30
30
20
10
Deleted 30
Top element is 20
```

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AI Help

HelloWorld.cpp

Stack Application (Infix to Postfix Conversion)

AI CPP RUN

```c
#include <stdio.h>
#include <ctype.h>
#define MAX 100

char stack[MAX];
int top = -1;

void push(char x) {
    stack[++top] = x;
}

char pop() {
    return stack[top--];
}

int priority(char x) {
    if (x == '(') return 0;
    if (x == '+' || x == '-') return 1;
    if (x == '*' || x == '/') return 2;
    return 0;
}

int main() {
    char exp[100];
    printf("Enter Infix Expression: ");
    scanf("%s", exp);

    for (int i = 0; exp[i] != '\0'; i++) {
        if (isalnum(exp[i]))
            printf("%c", exp[i]);
```

STDIN

5

Output: 2 ms | 4.5MB

```
Enter Infix Expression: 5
```

HelloWorld.cpp

Queue Operations (ENQUEUE, DEQUEUE, DISPLAY)

AI CPP RUN

```c
#include <stdio.h>
#define MAX 5

int queue[MAX], front = -1, rear = -1;

void enqueue(int value) {
    if (rear == MAX - 1)
        printf("Queue Overflow\n");
    else {
        if (front == -1) front = 0;
        queue[++rear] = value;
        printf("Inserted %d\n", value);
    }
}

void dequeue() {
    if (front == -1 || front > rear)
        printf("Queue Underflow\n");
    else
        printf("Deleted %d\n", queue[front++]);
}

void display() {
    if (front == -1)
        printf("Queue is Empty\n");
    else {
        for (int i = front; i <= rear; i++)
            printf("%d ", queue[i]);
        printf("\n");
    }
```

STDIN

Input for the program (Optional)

Output: 1 ms | 4.5 MB

```
Inserted 10
Inserted 20
Inserted 30
10 20 30
Deleted 10
20 30
```

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HelloWorld.cpp

Tree Traversals (Inorder, Preorder, Postorder)

AI CPP RUN

```c
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node *left, *right;
};

struct Node* create(int value) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = value;
    newNode->left = newNode->right = NULL;
    return newNode;
}

void inorder(struct Node* root) {
    if (root != NULL) {
        inorder(root->left);
        printf("%d ", root->data);
        inorder(root->right);
    }
}

void preorder(struct Node* root) {
    if (root != NULL) {
        printf("%d ", root->data);
        preorder(root->left);
        preorder(root->right);
    }
}
```

STDIN

Input for the program (Optional)

Output: 2 ms | 4.6 MB

```
Inorder: 4 2 5 1 3
Preorder: 1 2 4 5 3
Postorder: 4 5 2 3 1
```

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HelloWorld.cpp

Hashing using Linear Probing

AI CPP RUN

```c
#include <stdio.h>
#define SIZE 10

int hashTable[SIZE];

void init() {
    for (int i = 0; i < SIZE; i++)
        hashTable[i] = -1;
}

void insert(int key) {
    int index = key % SIZE;

    while (hashTable[index] != -1) {
        index = (index + 1) % SIZE;
    }

    hashTable[index] = key;
}

void display() {
    for (int i = 0; i < SIZE; i++) {
        if (hashTable[i] != -1)
            printf("%d -> %d\n", i, hashTable[i]);
        else
            printf("%d -> Empty\n", i);
    }
}

int main() {
```

STDIN

Input for the program (Optional)

Output:

2 ms | 4.5 MB

```
0 -> 10
1 -> 20
2 -> Empty
3 -> Empty
4 -> Empty
5 -> 15
6 -> Empty
7 -> 7
8 -> Empty
9 -> Empty
```